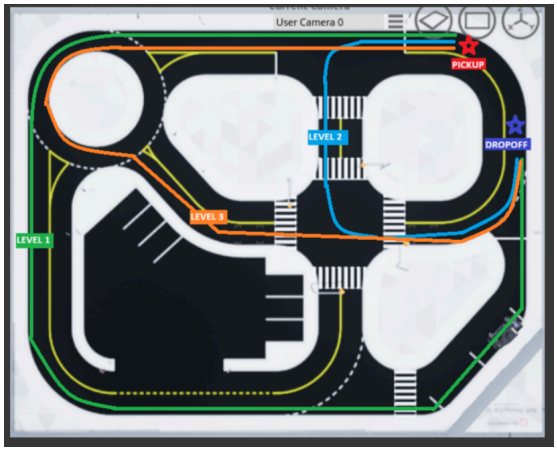
UWAFT Software Challenge:

Students are an autonomous taxi company and are tasked with picking up and dropping off a passenger. There are 3 levels they can choose from:

1. Simple route with a stop sign
2. Stop sign, one stop light
3. Stop sign, one stop light with a pedestrian



Goal: Provide students with the tools and activities to get them comfortable using Matlab/ Simulink/ Quanser! We want them to know where to put the tools!

You will not be scored differently based on what levels

Topic	Resources
Stop sign detection	1. Example network(s) and workflow 2. Example dataset and workflow for training an ACF detector 3. Example dataset and workflow for training R-CNN Detector
Obstacle Detection	1. Obstacle Avoidance Using a Camera Sensor 2. Ground Plane and Obstacle Detection Using Lidar 3. Obstacle Avoidance Using Adaptive Model Predictive Control
Moving object detection	1. Tracking and Motion Estimation Documentation 2. Detect and Track Vehicles Using Lidar Data (MATLAB and Simulink versions)

Traffic Light detection	1. Model Traffic Light Decision Logic (not detection, but relevant)
Speed controls	
Overall Algorithm and architecture	1. Object detection and classification using deep learning and LIDAR data 2. Path planning using Pure Pursuit algorithm 3. Build occupancy maps using LIDAR data
Additional resources	1. Collection of perception-related video tutorials 2. Collection of pre-trained detection networks 3. Configuring Simulation settings in Simulink 4. Automated parking valet example 5. IMPORTANT - Coordinate systems in Automated Driving Toolbox